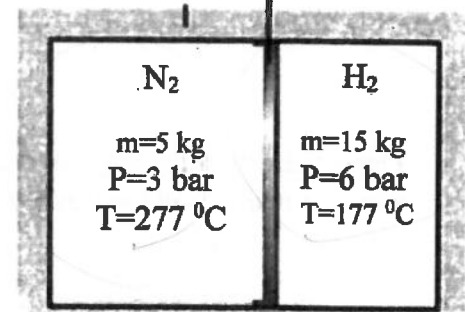


(12 points)-Five kg of Nitrogen (N_2) initially at 3.0 bar, 277 °C and fifteen kg of Hydrogen (H_2) initially at 6 bar, 177 °C are confined to opposite sides of a rigid container by a partition as shown in figure below. The partition is free to move and allows mixing from one gas to the other while receiving energy by heat transfer from the surroundings. The final equilibrium temperature is 207 °C. Assuming that no energy is transferred to, or from, the gas by work and no change in kinetic or potential energy. Using the Ideal Gas Specific Heat Table at constant C_v . (Show all steps of

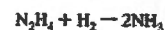
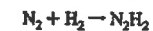
your summary in energy balance equation). ($\bar{R} = 8314 \frac{N.m}{kmol.k}$) = ($\bar{R} = 8.314 \frac{kJ}{kmol.k}$).

Determine the following: N_2 $m_1 = 5 kg$ $P_1 = 3 bar$ $T_1 = 277^\circ C$ $550 K$ $T_f = 207^\circ C$ $480 K$
 H_2 $m_2 = 15 kg$ $P_2 = 6 bar$ $T_2 = 177^\circ C$ $450 K$

- a. (6)-What is the final pressure after complete mixing in the state of equilibrium in bar? (You are NOT required to know the chemical reactions!)



The reaction of N_2 fixation and its reduction to NH_3 is thought to take place in three consecutive steps:



$$P_f V_f = m_f R T_f \rightarrow V_1 = \frac{m_1 R T_1}{P_1} + V_2 = \frac{m_2 R T_2}{P_2}$$

$$P_f = \frac{(m_1 + m_2) R T_f}{\frac{m_1 R T_1}{P_1} + \frac{m_2 R T_2}{P_2}} = \frac{(m_1 + m_2) T_f}{\frac{m_1 T_1}{P_1} + \frac{m_2 T_2}{P_2}}$$

$$P_f = \frac{(5+15) kg (207+273) K}{\frac{5(277+273) K}{3 bar} + \frac{15 kg (177+273) K}{6 bar}} = \frac{9600 K \cdot kg}{2041.66 \frac{K \cdot kg}{bar}} = 4.702 bar$$

$$P_f = 4.702 bar$$

1. (10)-Determine the specific volume (v), in ft^3/lb_m , of Sulfur dioxide (SO_2) at 760.7 psi and 353.75 °F, using the attached Generalized Compressibility Chart.

$$(\bar{R} = 1.986 \frac{\text{Btu}}{\text{lbmol} \cdot ^\circ\text{R}} = 1545 \frac{\text{ft} \cdot \text{lb}_f}{\text{lbmol} \cdot ^\circ\text{R}})$$

$$p_R = p/p_c$$

$$T_R = T/T_c$$

(Please mark your work on attached chart to identify Z).

(Show the full details of your calculation here)! Carry the given system of units!

given: $\text{SO}_2 \Rightarrow \text{gas}$

from table:

$$P = 760.7 \text{ psi} \left(\frac{1 \text{ atm}}{14.7 \text{ psi}} \right) = 51.75 \text{ atm}$$

$$P_c = 77.7 \text{ atm}$$

$$T = 353.75^\circ\text{F} (+ 459.67^\circ\text{R}) = 813.42^\circ\text{R}$$

$$T_c = 775^\circ\text{R}$$

$$P_R = \frac{P}{P_c} = \frac{51.75 \text{ atm}}{77.7 \text{ atm}} = 0.666$$

$$M = 64.06 \text{ lb/lbmol}$$

$$T_R = \frac{T}{T_c} = \frac{813.42^\circ\text{R}}{775^\circ\text{R}} = 1.0496$$

$$Z = 0.78$$

$$PV = ZRT \Rightarrow v = \frac{Z(\bar{R})T}{P}$$

$$v = \frac{(0.78) \left(\frac{1545 \text{ ft} \cdot \text{lb}_f/\text{lbmol} \cdot ^\circ\text{R}}{64.06 \text{ lb/lbmol}} \right) (813.42^\circ\text{R})}{(760.7 \text{ lbf/in}^2) \left(\frac{1 \text{ in}}{1 \text{ ft}} \right)^2}$$

$$v = 0.1397 \text{ ft}^3/\text{lb}_m$$

$$PV = ZRT$$

$$R = \frac{\bar{R}}{M} \text{ and } v = \text{specific volume}$$

$$Z = \text{Compressibility factor}$$

Atomic or Molecular Weights and Critical Properties of Some Selected Elements and Compounds

Substance	Chemical Formula	M (lb/lbmol)	T_c (°R)	p_c (atm)	$Z_c = \frac{p_c v_c}{RT_c}$
Propylene	C_3H_6	42.08	657	45.6	0.276
Refrigerant 12	CCl_2F_2	120.92	693	40.6	0.278
Refrigerant 22	CHClF_2	86.48	665	49.1	0.267
Refrigerant 134a	$\text{CF}_3\text{CH}_2\text{F}$	102.03	673	40.2	0.260
Sulfur dioxide	SO_2	64.06	775	77.7	0.268
Water	H_2O	18.02	1165	218.0	0.233

$$\Delta U = -W$$

2. (10)-A well-insulated rigid tank having a **constant volume** of 25 ft³ contains **saturated water vapor** at 270 °F. The water is rapidly stirred until the pressure is 80 lbf/in². Using the Table Properties of Saturated Water (Liquid + Vapor) and Table of Superheated Water Vapor, determine:

given:

$$V = 25 \text{ ft}^3$$

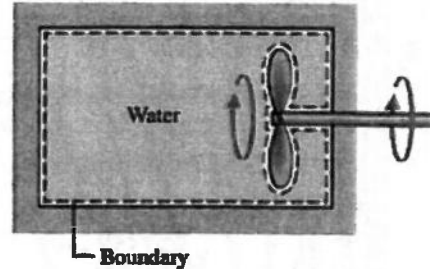
$$T_1 = 270^\circ \text{F}$$

$$P_2 = 80 \text{ lbf/in}^2$$

$$\Delta KE + \Delta PE + \Delta U = Q - W$$

(Tables are on P. 7)!

Carry the given system of units!



- a. (2 points)-The value of pressure (P_1), specific volume (v_1) and internal energy (u_1) at initial state?

$$P_1 = 41.85 \text{ lbf/in}^2$$

$$v_1 = 10.07 \text{ ft}^3/\text{lb}$$

$$u_1 = 1093 \text{ BTU/lb}$$

- b. (2 points)-The mass of water vapor at initial state in lb_m?

$$\begin{aligned} m &= \frac{V}{v} \\ &= \frac{25 \text{ ft}^3}{10.07 \text{ ft}^3/\text{lb}_m} \\ &= 2.48 \text{ lb}_m \end{aligned}$$

- c. (2 points)-The final temperatures T_2 and internal energy (u_2) in final state? (No interpolation)! $v_1 = v_2$ @ $P_2 = 80 \text{ lbf/in}^2$

$$T_2 = 900^\circ \text{F}$$

$$u_2 = 1332 \text{ BTU/lb}$$

- d. (4 points)-The work during the process, in BTU?

$$\Delta KE + \Delta PE + \Delta U = Q - W$$

$$\Delta U = -W$$

$$\Delta U = m(u_2 - u_1)$$

$$= (2.48 \text{ lb}_m)(1332 \text{ BTU/lb}_m - 1093 \text{ BTU/lb}_m)$$

$$= 592.72 \text{ BTU}$$

$$W = -592.72 \text{ BTU}$$

5. (6)-The 500-liters (Five hundred liters) rigid tank contains Carbon dioxide (CO₂) at 277 °C and 1600 kPa. What is the mass of gas in kg by using Ideal Gas Law?

$$101.3 \text{ kPa} =$$

$$1.013 \text{ bar}$$

$$1 \text{ kPa} = .01 \text{ bar}$$

$$(\bar{R} = 8.314 \frac{\text{kJ}}{\text{Kmol.K}} = 8314 \frac{\text{J}}{\text{Kmol.K}})$$

$$3.785 \text{ L} = .003785 \text{ m}^3$$

$$PV = mRT$$

$$V = 500 \text{ L} = 0.5 \text{ m}^3$$

$$T = 277^\circ\text{C} = 550 \text{ K}$$

$$M = 44.01 \text{ kg/kmol}$$

$$P = 1600 \text{ kPa} = 1,600,000 \text{ Pa} \\ = 16 \text{ bar}$$

$$m = \frac{PV}{RT}$$

$$m = \frac{16(10^5) \frac{\text{N}}{\text{m}^2} (.5) \text{ m}^3}{8314 \frac{\text{Nm}}{\text{Kmol.K}} (550) \text{ K} \left(\frac{1 \text{ kmol}}{44.01 \text{ kg}} \right)}$$

$$m = 7.6996 \text{ kg}$$